

# Snake App Summary

Repo evidence only: `index.html`, `styles.css`, `js/game.js`, and `js/main.js`

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## WHAT IT IS

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Snake is a static browser game that recreates the classic arcade experience in plain HTML, CSS, and JavaScript. The entire UI and gameplay loop run in the page, and no backend, package manifest, or build tooling was found in the repo.

## WHO IT IS FOR

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Players who want a lightweight browser game with both keyboard and on-screen controls, including smaller-screen use suggested by the responsive layout.

## WHAT IT DOES

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- Renders a 16 x 16 board and paints snake, head, and food states directly in the DOM.
- Starts movement from Arrow keys or WASD and blocks reverse-direction turns.
- Shows a live score in the header while updating status text for start, pause, play, win, and game-over states.
- Includes touch-friendly directional buttons for up, left, down, and right input.
- Lets the player pause and resume with the button or Space key.
- Lets the player restart with the button or the R key.
- Randomly places food, grows the snake on pickup, detects collisions, and declares a win when the board is full.

## HOW IT WORKS

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- `index.html` defines the shell: title, score panel, status message, board container, action buttons, touch controls, and script loading order.
- `js/game.js` exposes `window.SnakeGame` and owns the game rules: initial state, seeded RNG, food placement, direction changes, pause logic, collisions, and tick-to-tick state transitions.
- `js/main.js` bootstraps the page, builds the board grid, maps keyboard and button input to game actions, advances the game every 140 ms, and re-renders score, status, and cell classes.
- `styles.css` supplies the visual system: centered panel layout, responsive stacking under 520 px, and color/state styling for the board, controls, and score panel.
- External services, APIs, persistence, analytics, and backend components: Not found in repo.

## HOW TO RUN

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- Open `index.html` in a browser.
- Use Arrow keys or WASD to start and steer; Space pauses and R restarts.
- Dedicated install, build, test, or server steps: Not found in repo.